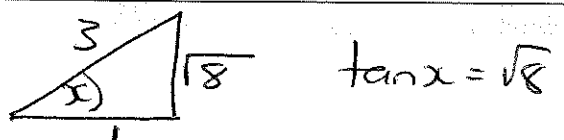


SECTION A [Multiple choice questions] - 18 marks

QUESTION 1

If $\cos x = -\frac{1}{3}$, $\pi < x < \frac{3\pi}{2}$, then $\tan x =$



- (a) $\frac{1}{\sqrt{8}}$ (b) $-\frac{1}{\sqrt{8}}$ (c) $\frac{3}{\sqrt{8}}$ (d) $-\frac{3}{\sqrt{8}}$ (e) $\sqrt{8}$ (f) $-\sqrt{8}$ (g) None of (a) - (f)

QUESTION 2

Consider the table below.

x	1	2	4	8
$f(x)$	8	4	3	6
$g(x)$	4	1	8	2

$$(g \circ f)(1) = g(f(1)) = g(8) = 2$$

$(g \circ f)(1) =$

- (a) 1 (b) 2 (c) 4 (d) 8 (e) 12 (f) 32 (g) None of (a) - (f)

QUESTION 3

Consider $f(t) = \ln(e^t - 1)$. The domain of f is:

$$e^t - 1 > 0 \Rightarrow e^t > 1 \Rightarrow \ln e^t > \ln 1 \Rightarrow t > 0$$

- (a) $[0, 1]$ (b) $(0, 1)$ (c) $[0, \infty)$ (d) $(0, \infty)$ (e) $[1, \infty)$ (f) $(1, \infty)$ (g) $\mathbb{R}$ (h) None of (a) - (g)

QUESTION 4

If $f(x) = \ln(2-x)$, then $f^{-1}(x) =$

$$y = \ln(2-x) \Rightarrow e^y = e^{\ln(2-x)} = 2-x \Rightarrow x = 2 - e^y$$

Swap x and y :
 $y = f^{-1}(x) = 2 - e^x$

- (a) e^{2-x} (b) e^{x-2} (c) $2 - e^x$ (d) $e^x - 2$ (e) $e^2 - x$ (f) $x - e^2$ (g) None of (a) - (f)

QUESTION 5

For which value(s) of x is $y = \tan^{-1}(3x-1)$ defined?

Arctan defined on $(-\infty, \infty)$
 $-\infty < 3x-1 < \infty \Rightarrow$

- (a) $x \in [-1, 1]$ (b) $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ (c) $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$ (d) $x \in [0, 1]$ (e) $x \in [0, \pi]$ (f) $x \in \mathbb{R}$

QUESTION 6

Choose the one single equation that is not equivalent to the rest of the equations, or choose (e):

(a) $e^{e^x} = 10$

(b) $x = \ln \ln(10)$

(c) $e^x = \ln 10$

(d) $e^{2x} = \ln 100$

(e) The equations in (a) - (d) are equivalent to each other.

(a) (b) and (c) are equivalent.

$$\text{If } e^x = \ln 10 \Rightarrow (e^x)^2 = e^{2x} = (\ln 10)^2 \neq \ln 100$$

$$7 - 1 < |3 - x|$$

$$\Rightarrow |3 - x| > 6$$

QUESTION 7

The solution set of $7 - |3 - x| < 1$ is:

(a) $(-9, 3)$	(b) $(-3, 9)$	(c) ϕ	(d) $(-\infty, -9) \cup (3, \infty)$	(e) $(-\infty, -3) \cup (9, \infty)$	(f) $\mathbb{R} \setminus 3$	(g) None of (a) - (f)
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QUESTION 8

Consider the following statements:

- (1) x^n is an exponential function.
- (2) x^{-n} is a reciprocal function.
- (3) $x^{\frac{1}{n}}$ is a root function.

See p. 27

See p. 32

See p. 32

Choose the correct answer:

- (a) Only statement (1) is correct.
- (b) Only statement (2) is correct.
- (c) Only statement (3) is correct.
- (d) Only statements (1) and (2) are correct.
- (e) Only statements (1) and (3) are correct.
- ☒ (f) Only statements (2) and (3) are correct.
- (g) Statements (1), (2) and (3) are correct.
- (h) Statements (1), (2) and (3) are false.

QUESTION 9

Consider the following statements:

- (1) $\frac{e^x}{e^y} = e^{\frac{x}{y}}$.
- (2) If $a > 0$, $e^x + a^x = (e + a)^x$.
- (3) If $a > 0$, $\frac{e^x}{a^x} = \left(\frac{e}{a}\right)^x$.

See: Laws of exponents,
p. 47.

Choose the correct answer:

- (a) Only statement (1) is correct.
- (b) Only statement (2) is correct.
- ☒ (c) Only statement (3) is correct.
- (d) Only statements (1) and (2) are correct.
- (e) Only statements (1) and (3) are correct.
- (f) Only statements (2) and (3) are correct.
- (g) Statements (1), (2) and (3) are correct.
- (h) Statements (1), (2) and (3) are false.

QUESTION 10

The domain of the function $f(x) = \frac{1 + \ln(x+1)}{\ln(x-3)}$ is

- (a) $(3, 4) \cup (4, \infty)$.
 (b) $(-1, 3) \cup (3, \infty)$.
 (c) $x \in \mathbb{R} \setminus \{3\}$.
 (d) $x \in \mathbb{R} \setminus \{-1 + \frac{1}{e}\}$.
 (e) $(-1, -1 + \frac{1}{e}) \cup (-1 + \frac{1}{e}, 3) \cup (3, \infty)$.
 (f) $(-1, \infty)$.
 (g) $(3, \infty)$.
 (h) None of these.

$\ln(x+1)$ defined if $x+1 > 0$
 $\Rightarrow x > -1$

$\therefore$ We have $x > -1$ and $x > 3$
 $\Rightarrow x > 3$

Also: $\ln(x-3) \neq 0$
 $\Rightarrow e^{\ln(x-3)} \neq e^0$

$\Rightarrow x-3 \neq 1$

$\Rightarrow x \neq 4$

$\Rightarrow x \in (3, 4) \cup (4, \infty)$

QUESTION 11

Consider the following statements:

- (1) The distance from the origin to x is less than a if and only if x lies between $-a$ and a .
 (2) $S \cup T$ is the set consisting of all elements that are in S and T .
 (3) $A \cap \phi = A$.

Choose the correct answer:

- (a) Only statement (1) is correct.
 (b) Only statement (2) is correct.
 (c) Only statement (3) is correct.
 (d) Only statements (1) and (2) are correct.
 (e) Only statements (1) and (3) are correct.
 (f) Only statements (2) and (3) are correct.
 (g) Statements (1), (2) and (3) are correct.
 (h) Statements (1), (2) and (3) are false.

QUESTION 12

The table below shows the position of a motorcyclist after acceleration from rest.

See Unit 2.1.

t (seconds)	0	1	2	3	4	5	6
s (meters)	0	1	5	12	23	38	52

Find the average velocity for the time period $[2, 5]$.

- (a) 3 (b) 5 (c) 8 (d) 11 (e) 33 (f) $\frac{37}{3}$ (g) None of (a) - (f)

SECTION B [Fill in questions] - 15 marks

SECTION B: Write only your final, fully simplified answer in the block provided. Only this will be marked.

QUESTION 13 [1 mark]

If $\frac{(e^x)^3(e^{3x})^2}{e^{4x}} = e^a$, then $a =$

See exponential ~~laws~~ Laws: P.

QUESTION 14 [1 mark]

Solve the inequality $\frac{|x-2|}{x-4} < 0$. Write down your answer using interval notation.

We know $|x-2| > 0$

$$\Rightarrow x-4 < 0$$

$$\Rightarrow x < 4$$

$$(-\infty, 2) \cup (2, 4)$$

But also:

$$x \neq 2$$

QUESTION 15 [1 mark]

Complete the definition of a one-to-one function:
A function f is called a one-to-one function if

see textbook

QUESTION 16 [1 mark]

Complete the definition of an odd function:
A function $f : A \rightarrow B$ is odd if

see textbook

QUESTION 17 [2 marks]

Write down the definition of the function $\cos^{-1}$ ($\arccos = \cos^{-1}$):

see textbook

$$81^t = (3^4)^t \\ = 3^{4t}$$

QUESTION 18 [2 marks]

If $3^a 3^b 3^c 3^d 3^e = 81^t$, find the average of a, b, c, d and e in terms of t .

$$4t/5$$

$$\Rightarrow a+b+c+d+e = 4t$$

Average of a, b, c, d, e is $\frac{a+b+c+d+e}{5} = \frac{4t}{5}$.

QUESTION 19 [2 marks]

By reflecting the graph of $y = -x^2 - 4x + 5$ about the x -axis, and shifting it one unit to the right, we obtain the graph of

$$x^2 + 2x - 8$$

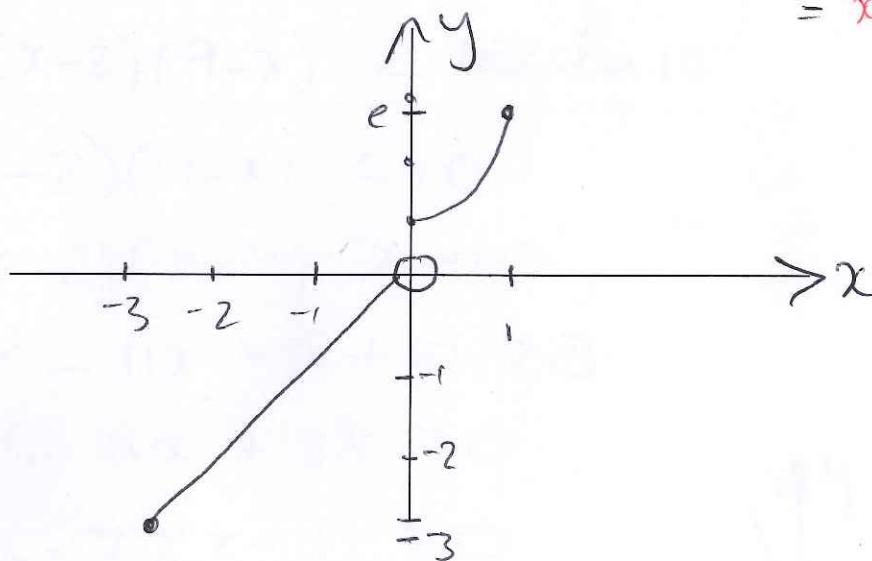
Reflect: $x^2 + 4x - 5$
Shift:

$$(x-1)^2 + 4(x-1) - 5 \\ = x^2 - 2x + 1 + 4x - 4 - 5 \\ = x^2 + 2x - 8$$

QUESTION 20 [5 marks]

Let f be the one-to-one function, defined by $f(x) = \begin{cases} -|x| & \text{if } -3 \leq x < 0, \\ e^x & \text{if } 0 \leq x \leq 1. \end{cases}$

(i) Sketch the graph of the function f on the axes provided, indicating all relevant intercepts and information. (2)



(ii) Determine the domain of f .

(1)

$$[-3, 1]$$

(iii) Determine the range of f .

(1)

$$[-3, 0) \cup [1, e]$$

(iii) Determine $f^{-1}(2)$.

$$\ln 2$$

End of Section B

Suppose ⁽¹⁾

$$f^{-1}(2) = y$$

$$\Rightarrow f(y) = 2$$

$$\Rightarrow e^y = 2$$

$$\Rightarrow y = \ln 2$$

SECTION C [Full written answers] - 17 marks

QUESTION 21 [6 marks]

Solve for x . Write your answer in interval notation.

(a) $\ln(x-2) + \ln(9-x) < \ln 10$

(4)

$$\begin{aligned}x-2 &> 0 \text{ and } 9-x > 0 \\ \Rightarrow x &> 2 \text{ and } x < 9 \\ \Rightarrow x &\in (2, 9)\end{aligned}$$

$$\ln(x-2) + \ln(9-x) < \ln 10$$

$$\Rightarrow \ln(x-2)(9-x) < \ln 10$$

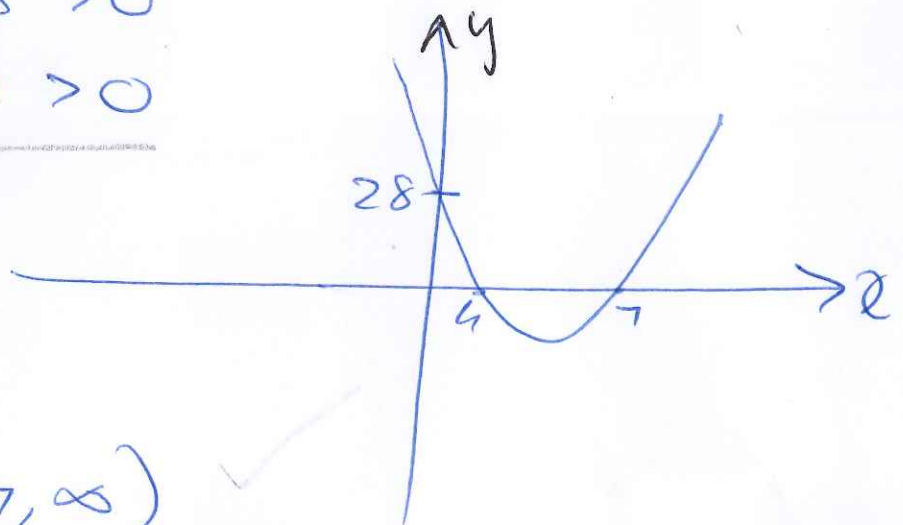
$$\Rightarrow (x-2)(9-x) < 10$$

$$\Rightarrow (x-2)(x-9) > -10$$

$$\Rightarrow x^2 - 11x + 18 + 10 > 0$$

$$\Rightarrow x^2 - 11x + 28 > 0$$

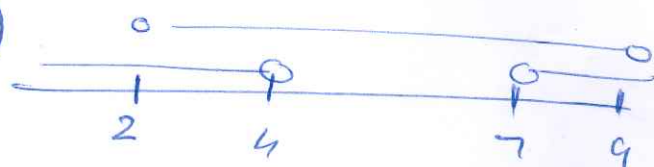
$$\Rightarrow (x-7)(x-4) > 0$$

From sketch:

$$x \in (-\infty, 4) \cup (7, \infty)$$

but we know: $x \in (2, 9)$

$$\Rightarrow \underline{x \in (2, 4) \cup (7, 9)}$$



(b) $6e^x > 5e^{-1}$

(2)

$$e^{x+1} > \frac{5}{6}$$

$$\ln e^{x+1} > \ln \frac{5}{6}$$

$$\Rightarrow x+1 > \ln \frac{5}{6}$$

$$\Rightarrow x > \ln \frac{5}{6} - 1$$

$$\Rightarrow x \in (\ln \frac{5}{6} - 1, \infty)$$

ROUGH WORKING

QUESTION 22 [4 marks]

- (i) Determine if the function
- f
- is even, odd, or neither even nor odd.

(2)

$$f(x) = |x| + \sin^2(x).$$

$$\begin{aligned} f(-x) &= |-x| + \sin^2(-x) \\ &= |x| + (\sin(-x))^2 \\ &= |x| + (-\sin x)^2 \\ &= |x| + \sin^2 x \\ &= f(x) \end{aligned}$$

$\Rightarrow f$ is an even function

- (ii) We refer to the definition of an odd function given in Question 16. Suppose
- g
- is an odd function, show that the composite function
- $g \circ g$
- is odd. Motivate your steps.

(2)

Given: g is odd, $\Rightarrow g(-x) = -g(x)$

$$\begin{aligned} (g \circ g)(-x) &= g(g(-x)) \\ &= g(-g(x)) \quad \text{since } g \text{ is odd} \\ &= -g(g(x)) \quad \text{since } g \text{ is odd} \\ &= -(g \circ g)(x) \end{aligned}$$

$\Rightarrow g \circ g$ is an odd function

QUESTION 23 [2 marks]Use the definition of the $\cos^{-1}$ function given in Question 17 and determine the value of

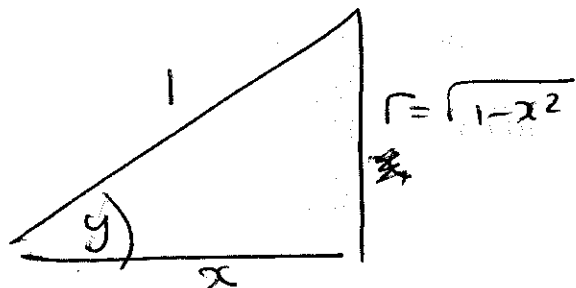
(3)

$$\sin(2 \cos^{-1} x).$$

Explain all your steps.

$$\text{Let } y = \cos^{-1} x$$

$$\Rightarrow \cos y = x \text{ and } y \in [0, \pi]$$



$$x^2 + r^2 = 1$$

$$\Rightarrow r^2 = 1 - x^2$$

$$\Rightarrow r = \sqrt{1-x^2}$$

$$\text{Now } \sin(2 \cos^{-1} x)$$

$$= \sin(2y)$$

$$= 2 \sin y \cos y$$

$$= 2 \sqrt{1-x^2} \cdot x$$

$$= \underline{2x \sqrt{1-x^2}}$$

$$\sin y = \frac{\sqrt{1-x^2}}{1}$$

QUESTION 24 [2 marks]

Solve for k if $\sqrt{2k+1} - \frac{24}{\sqrt{2k+1}} = 5$.

Let $y = \sqrt{2k+1}$, therefore: $y > 0$

$$\Rightarrow y - \frac{24}{y} = 5$$

$$\Rightarrow y^2 - 24 = 5y$$

$$\Rightarrow y^2 - 5y - 24 = 0$$

$$\Rightarrow (y-8)(y+3) = 0$$

$$\Rightarrow y = 8 \text{ or } y = -3$$

N/A, since $y > 0$

$$\Rightarrow \sqrt{2k+1} = 8$$

$$\Rightarrow 2k+1 = 64$$

$$\Rightarrow 2k = 63$$

$$\Rightarrow \underline{k = \frac{63}{2}}$$

QUESTION 25 [3 marks]

Consider the function

$$f(x) = 1 + \frac{1}{x}.$$

Find the inverse function, f^{-1} , together with the domain and range of f^{-1} . Use interval notation.

$$\text{Let } y = 1 + \frac{1}{x}$$

$$\text{then } y - 1 = \frac{1}{x}$$

$$\text{and } x = \frac{1}{y-1}$$

Now, interchange y and x :

$$y = \frac{1}{x-1}$$

$$\Rightarrow f^{-1}(x) = \frac{1}{x-1}$$

and

$$D_f = (-\infty, 0) \cup (0, \infty) = R_{f^{-1}}$$

$$D_{f^{-1}} = (-\infty, 1) \cup (1, \infty)$$